GBU 2301: Business Statistics

Practice Questions: Chapters 1–4

Fall 2026 | Dr. Ömer ÇAYIRLI

With solutions

Chapter 1: Data and Statistics

Questions 1–6.

Support each answer with the information in the case; definitions alone are not sufficient.

Question 1 (Easy)

Alpha Repairs records parts used and the elapsed completion time of each job. Each listed part was used once.

Job	Part used	Job completion time (minutes)
J01	Battery	30
J01	Cable	30
J01	Connector	30
J02	Screen	90

a. Construct a table suitable for reporting average completion time per job. State what one observation represents and explain the effect of averaging the original rows. No calculation is required.

b. Can the original register be used directly to count each type of part used? Explain.

Question 1: Solution

a.

Job	Completion time (minutes)
J01	30
J02	90

One observation represents one repair job. The original register gives J01 three rows and J02 one row. Averaging those rows would give the shorter job three times the weight and understate the average completion time per job.

b. Yes. Each row records one part used once, so the original register can be used directly to count parts. Battery, cable, connector, and screen each appear once.

The appropriate observation depends on the question: a job for completion times, a part used for part counts.

Question 2 (Easy)

Beta Service Centres records these variables:

Variable	Example value	Meaning
centre_code	104	Identifier assigned to the centre
service_rating	4	Customer chooses one category: very poor = 1, poor = 2, fair = 3, good = 4, very good = 5
complaint_count	8	Number of complaints received that day
waiting_minutes	12.5	Measured customer waiting time
room_temperature_c	20	Room temperature in degrees Celsius

a. Classify each variable in a table: categorical or quantitative, discrete or continuous where applicable, and measurement scale.

b. Which statements below can be defended from the stated measurements? Explain each decision.

Statement	Comparison
A	A centre at 20°C is 5°C warmer than one at 15°C.
B	A rating of 4 indicates twice the service quality of a rating of 2.
C	A waiting time of 25 minutes is twice a waiting time of 12.5 minutes.
D	Centre 104 has higher service quality than Centre 102.

Question 2: Solution

a.

Variable	Statistical type	Measurement scale
centre_code	Categorical	Nominal
service_rating	Categorical, ordered	Ordinal
complaint_count	Quantitative, discrete	Ratio
waiting_minutes	Quantitative, continuous	Ratio
room_temperature_c	Quantitative, continuous	Interval

A code identifies an individual centre. A rating identifies an ordered category. Neither becomes quantitative merely because it is stored as a number.

Question 2: Solution

b.

Statement	Assessment
A	Supported. Differences on the Celsius scale are meaningful, although ratios of Celsius values are not.
B	Not supported. The rating categories do not establish equal numerical distances or meaningful ratios of service quality.
C	Supported. Waiting time is a duration with a meaningful zero, so the ratio is meaningful.
D	Not supported. Centre codes identify centres; their numerical order does not rank service quality.

Question 3 (Average)

Gamma Retail records monthly sales at two kiosks:

Kiosk	Month	Sales (thousand MAD)
Alpha	January	40
Alpha	February	60
Alpha	March	80
Beta	January	80
Beta	February	60
Beta	March	40

Both kiosks have average monthly sales of 60 thousand MAD over these three months. An analyst proposes replacing the six records with those two averages for the following reports:

Report	Purpose
A	Compare total sales contributed by each kiosk during January–March
B	Compare the kiosks' sales in March
C	Examine each kiosk's sales pattern when deciding April stock allocations

a. For which reports would the two averages retain the information needed for the stated sales comparison? For the others, specify the records you would use and identify their data structure.

Question 3 (Average)

b. State what one observation represents in the original table and name the structure of that dataset.

c. Management proposes equal April stock allocations because the averages are equal. Give a recommendation supported by the available records.

Question 3: Solution

a.

Report	Appropriate records and assessment
A	The averages are sufficient to compare January–March sales totals because both cover the same three months. Equal averages imply equal totals here. The summary is a cross-sectional comparison of kiosks over a common period.
B	Retain Alpha and Beta's March records. These form a March cross-section. The two period averages do not preserve that month's comparison.
C	Retain January, February, and March for each kiosk. Each kiosk's records form a time series; keeping both series together gives panel data.

b. One observation is a kiosk in a particular month. The original dataset is panel, or longitudinal, data because the same kiosks are observed repeatedly.

c. Equal April allocations are not justified by the averages alone. Alpha's sales rise from 40 to 80 thousand MAD, while Beta's fall from 80 to 40 thousand MAD.

Retain the monthly records and investigate current demand, product mix, stock already available, and any stock shortages before allocating stock. The patterns warrant different consideration, but three months of sales do not establish future demand or the appropriate allocation.

Question 4 (Average)

Delta Transport wants to understand satisfaction among all individual passengers who used its intercity bus services during September.

Survey information	Detail
Invitation list	6,000 September passengers who booked through the mobile app
Invitation procedure	Every passenger on the list was invited once
Responses	1,200 passengers, with one response per passenger
Result	74% of respondents selected satisfied or very satisfied
Other booking channels	Website, ticket offices, and travel agents

The survey team makes three statements:

1. Every passenger on our invitation list was invited.
2. We have a census of September passengers because every received response was analysed.
3. Passenger satisfaction in September was 74%.

a. Which statements can be retained in the public report, and which need revision? Give suitable wording where needed and identify the population actually described by the 74% figure.

b. Management offers funding to obtain twice as many responses next time. Recommend how the next survey should be organised and explain whether the larger response count would, by itself, support Statement 3.

Question 4: Solution

a.

Statement	Assessment
1	Retain. It accurately describes the invitation procedure, but says nothing about passengers outside the list or those who did not respond.
2	Revise. The target population is all individual September passengers. The 1,200 respondents are a sample, not a census of that population.
3	Revise to: “Among the 1,200 app-booking passengers who responded, 74% selected satisfied or very satisfied.”

The 74% figure describes the respondent group. The invitation list excludes passengers who booked only through other channels; within the list, those who responded may differ from those who did not.

b. Build a passenger list covering all booking channels and remove duplicate passenger entries. Draw a random sample from that list, with coverage of the relevant routes and departure times. Use consistent questions and follow up with selected passengers who do not respond.

Doubling responses from the same restricted invitation method would not resolve the missing coverage or necessarily resolve response differences. Random selection supports broader inference, but coverage and nonresponse still need attention.

Question 5 (Average)

Epsilon Retail compares two website checkout designs.

Feature	Study A	Study B
Participants	2,400 customers randomly selected from the current customer list; all participate	200 loyalty-club members who volunteer
Design received	Each customer chooses standard or simplified checkout	Each participant is randomly assigned standard or simplified checkout
Prices and delivery terms	The same under both designs	The same under both designs
Period	Both designs operate during the same month	Both designs operate during the same month
Recorded outcome	Whether the customer completes a purchase	Whether the customer completes a purchase

Each customer contributes one outcome. In both studies, the simplified checkout has the higher observed purchase-completion rate.

The marketing director prefers Study A because it includes more customers and they were randomly selected. The website manager prefers Study B because the design was randomly assigned and recommends applying its result to all customers.

a. Classify the studies and assess each manager's reasoning.

b. Which study would you rely on more for evidence that the design itself affects purchase completion? What limits remain when deciding whether to introduce it for all customers?

c. Describe a study that combines the useful features of these two designs.

Question 5: Solution

a.

Study	Classification	Strength and limitation
A	Observational: customers choose their design	Random selection supports representation of the current customer list. However, customers choosing the two designs may differ in purchase intention, experience, or device. Neither random selection nor the larger sample removes that problem.
B	Experimental: the design is randomly assigned	Random assignment strengthens the comparison between designs among participants. The volunteer loyalty-club sample does not establish representation of all customers.

Each manager identifies a genuine strength but overlooks a different limitation.

b. Study B provides stronger causal evidence because the design is assigned rather than chosen. Random assignment reduces systematic differences between the groups receiving the two designs; it does not guarantee exact balance.

The observed difference may still reflect chance variation. Its size and uncertainty have not been reported. Loyalty-club volunteers may also respond differently from other customers. Neither study alone establishes the benefit of a general rollout.

c. Randomly select participants from a customer list covering the intended users, then randomly assign checkout designs within that sample. Keep the other relevant conditions comparable and record purchase completion consistently.

Question 5: Solution

Selection addresses who is represented; assignment addresses the comparison between designs. Coverage, nonresponse, and changes in operating conditions still require attention.

Question 6 (Hard)

Zeta Manufacturing uses scheduled machine inspections. When an inspection detects wear, components can be replaced before a breakdown occurs.

The company is considering a model that predicts whether a machine will break down in the following month.

Project information	Detail
Records	10,000 machine-months from two recently opened plants, collected under the existing inspection schedule
Other plants	Older plants keep paper maintenance records, which were not included
Model development	Twelve candidate models were developed and compared using the 10,000 records
Reported result	The selected model correctly classifies 96% of those same records
Outcomes in the reported data	94% of machine-months had no breakdown
Exclusions before development	Some machine-months with confirmed breakdowns were labelled “exceptional operation” and left out; no further information about those cases is supplied

Management proposes inspecting machines predicted to have no breakdown once a month instead of twice, at every plant. Machines predicted to break down would retain twice-monthly inspections.

Recommend whether and how the company should proceed. Support your recommendation with the report and specify what evidence would change your decision.

Question 6: Solution

Retain the current inspection schedule while evaluating the model and the proposed policy. The report supports further investigation, not the proposed rollout.

Evidence	Assessment
96% accuracy; 94% of records have no breakdown	The model exceeds the accuracy of always predicting no breakdown. However, overall accuracy does not reveal how many breakdowns it misses or whether the improvement justifies changing inspections.
The same records are used for development, selection, and reported evaluation	Performance may be optimistic after selecting the best of twelve models. Independent evaluation is needed.
Some confirmed breakdown cases were excluded	Establish why they were excluded and whether those conditions can occur in the intended use. Genuine relevant breakdowns should not be removed simply because they are unusual.
Only two newer plants supply the records	Reliability at older plants has not been demonstrated. Their maintenance records need to be assessed and made comparable where possible.
Inspections can trigger replacement before failure	The recorded outcomes reflect the existing inspection schedule. Reducing inspections may increase breakdown risk, so accuracy under the old schedule does not establish performance under the proposed one.

First evaluate the selected model on separate, previously unused records covering the intended plants and operating conditions. Report correct and missed breakdown predictions, false alarms, and performance relative to the no-breakdown benchmark.

Question 6: Solution

If that evidence is satisfactory, test a limited policy change under a controlled pilot with safety safeguards. Compare breakdowns, downtime, maintenance costs, and inspection savings with a group retaining the current schedule.

Evidence of reliable prediction alone would not justify rollout. Evidence that the changed inspection policy maintains acceptable safety and operating outcomes at worthwhile cost would support proceeding.

Chapter 2: Tabular and Graphical Displays

Questions 7–16.

Report percentages to two decimal places when needed. Retain unrounded values during calculations.

Question 7 (Easy)

Eta Café records the payment method for twelve transactions. Each transaction uses one payment method.

Transaction	Method	Transaction	Method
1	Cash	7	Cash
2	Card	8	Card
3	Mobile	9	Cash
4	Card	10	Mobile
5	Cash	11	Card
6	Mobile	12	Cash

a. Construct a frequency table showing frequency, relative frequency, and percent frequency.

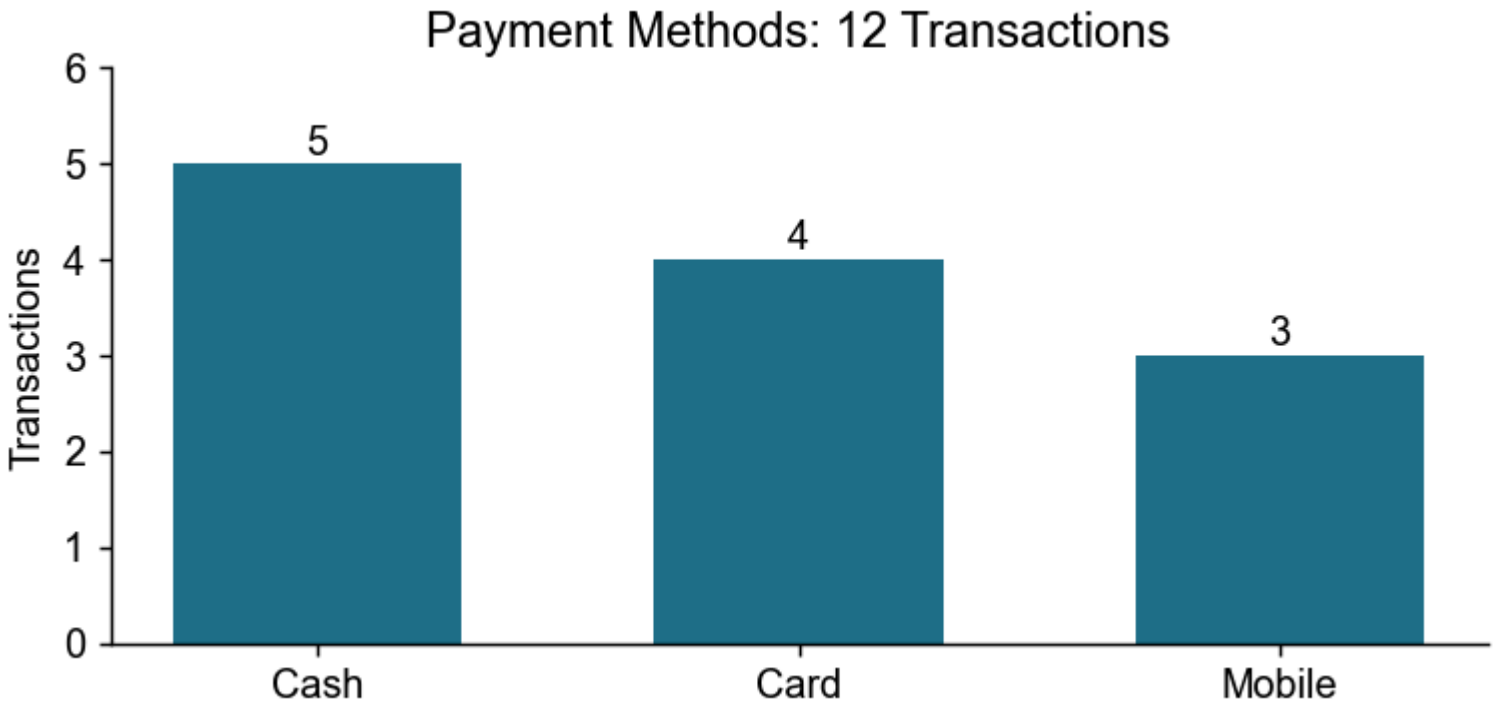
b. Sketch a bar chart of transaction counts, ordering the categories from most to least frequent. Include a title and a labelled numerical axis.

Question 7: Solution

$$\text{Relative frequency} = \frac{f}{12}, \qquad \text{Percent frequency} = 100\frac{f}{12}$$

Method	Frequency	Relative frequency	Percent frequency
Cash	5	$5/12 = 0.4167$	41.67%
Card	4	$4/12 = 0.3333$	33.33%
Mobile	3	$3/12 = 0.2500$	25.00%
Total	12	1.0000	100.00%

Question 7: Solution



The bars are separated because the payment methods are categories. Their lengths start from zero.

Question 8 (Easy)

Theta Appliances reports sales and returns for two stores. Each sale is one appliance, and each returned appliance is counted once. All sales have completed the same return window.

Store	Appliances sold	Appliances returned
Alpha	600	18
Beta	240	12

a. Calculate each store's return rate per 100 appliances sold and its share of all returned appliances.

b. Which store contributes more returned appliances, and which has the higher return rate? Explain why the rankings differ.

Question 8: Solution

a.

$$\text{Total returns} = 18 + 12 = 30$$

Measure	Alpha	Beta
Returns per 100 sold	$100(18/600) = 3.00$	$100(12/240) = 5.00$
Share of all returns	$100(18/30) = 60.00\%$	$100(12/30) = 40.00\%$

b. Alpha contributes more returned appliances: 18 rather than 12, or 60% of all returns. Beta has the higher return rate: 5 per 100 sold rather than 3.

Alpha sells more appliances, so it can have more returns despite a lower return rate. Counts describe the returns workload; rates describe the incidence of returns relative to sales.

Question 9 (Easy)

Iota Couriers records the following delivery times:

Delivery	1	2	3	4	5	6	7	8	9	10
Time (minutes)	47	32	41	38	32	55	46	41	60	39

a. Construct a stem-and-leaf display with a leaf unit of one minute. Include a key.

b. Use the display to determine the number and percentage of deliveries taking at least 45 minutes. Identify any repeated times.

Question 9: Solution

a.

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Stem | Leaves
  3  | 2 2 8 9
  4  | 1 1 6 7
  5  | 5
  6  | 0
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Key: $4 \mid 6$ represents 46 minutes; leaf unit = 1 minute.

b. The qualifying times are 46, 47, 55, and 60 minutes.

$$\text{Number} = 4, \qquad \text{Percentage} = 100\left(\frac{4}{10}\right) = 40\%$$

The repeated times are 32 and 41 minutes, each occurring twice.

Question 10 (Average)

Kappa Services is assembling a management report. The proposed displays are listed below.

Report purpose	Available data	Proposed display
Compare payment-method shares for the quarter	Four payment categories	Pie chart
Examine the distribution of customer waiting times	Individual waiting times in minutes	Bar chart with one bar for each customer
Compare booking-channel preferences at two branches	Channel counts; 240 customers at one branch and 80 at the other	Stacked bars of customer counts
Examine the relationship between advertising expenditure and orders	One expenditure-order pair for each of eight stores	Two separate pie charts
Follow monthly sales during the year	Twelve monthly sales totals	Line chart in calendar order

Which proposed displays would you retain, replace, or supplement? For each, specify the display you would use and explain how it serves the stated purpose. More than one choice can be defensible if justified.

Question 10: Solution

Purpose	Decision and justification
Payment-method shares	A pie chart can be retained for a simple composition of four categories. A bar chart is preferable if precise comparison between similar shares is important.
Waiting-time distribution	Replace with a histogram or, for a small dataset, a dot plot. Use waiting time on the horizontal axis and frequency on the vertical axis. Separate bars for individual customers obscure the distribution.
Booking-channel preferences	Use side-by-side percentage bars or 100% stacked bars, with each branch's percentages based on its own customer total. Report the totals of 240 and 80 alongside the display. Stacked counts can supplement this if workload also matters.
Advertising and orders	Replace with a scatter diagram: expenditure on the horizontal axis and orders on the vertical axis. Each store contributes one paired point. Separate pies do not show the paired relationship.
Monthly sales	Retain the line chart, with month on the horizontal axis and sales on the vertical axis. Calendar order preserves the sequence.

The display should answer the stated question. There is no requirement to use the same chart type throughout the report.

Question 11 (Average)

Lambda Logistics records loading times for twelve deliveries.

Delivery	1	2	3	4	5	6	7	8	9	10	11	12
Time (minutes)	18	7	30	12	25	9	34	20	3	29	10	23

a. Calculate the approximate class width for four classes using the observed range.

b. For the report, use four classes of width 10 minutes beginning at 0. Include the lower boundary and exclude the upper boundary of each class. Construct a table containing class midpoints, frequencies, percent frequencies, and cumulative percent frequencies.

c. Sketch the frequency histogram. How many deliveries took less than 20 minutes?

Question 11: Solution

a.

$$\text{Approximate class width} = \frac{34 - 3}{4} = 7.75 \text{ minutes}$$

The specified width of 10 gives convenient boundaries and covers all observations.

b.

Loading time, x	Midpoint	Frequency	Percent frequency	Cumulative percent
$0 \le x < 10$	5	3	25.00%	25.00%
$10 \le x < 20$	15	3	25.00%	50.00%
$20 \le x < 30$	25	4	33.33%	83.33%
$30 \le x < 40$	35	2	16.67%	100.00%
Total		12	100.00%	

For example:

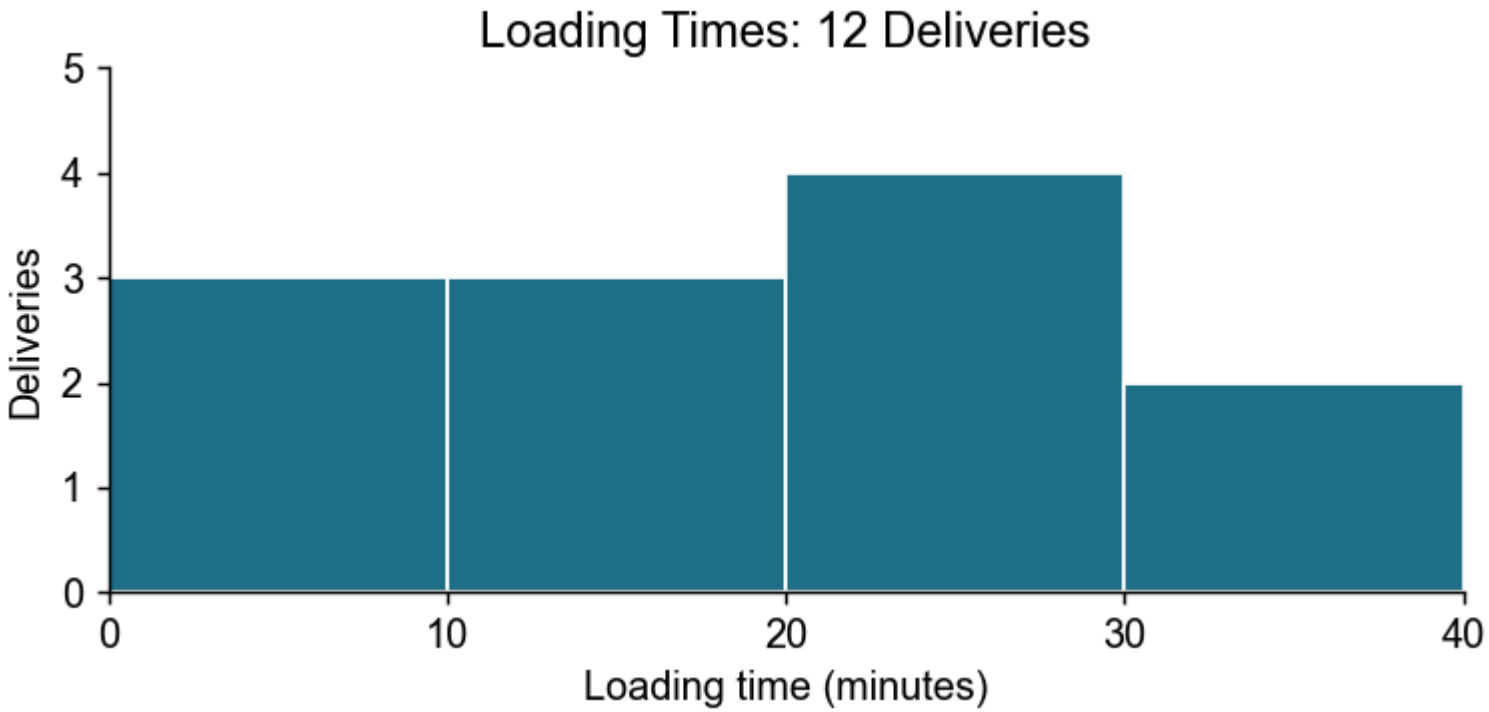
$$\text{Midpoint of } [20, 30) = \frac{20 + 30}{2} = 25$$

$$\text{Percent frequency} = 100\left(\frac{4}{12}\right) \approx 33.33\%$$

Question 11: Solution

The observations 10, 20, and 30 enter the classes beginning at those values.

c.



$$\text{Deliveries below 20 minutes} = 3 + 3 = 6$$

Question 12 (Average)

Mu Support retains only the following cumulative summary for twenty nonnegative customer waiting times. All waiting times were below 20 minutes.

Waiting-time threshold	Number below the threshold
5 minutes	4
10 minutes	11
15 minutes	17
20 minutes	20

a. Recover the frequencies and percent frequencies for the intervals $[0, 5)$, $[5, 10)$, $[10, 15)$, and $[15, 20)$.

b. The service target requires at least 80% of customers to wait less than 12 minutes. Determine whether the retained summary establishes that the target was met. Support your answer with the smallest and largest possible qualifying counts consistent with the table.

Question 12: Solution

a.

Waiting time (minutes)	Frequency	Percent frequency
$0 \le x < 5$	4	$100(4/20) = 20\%$
$5 \le x < 10$	$11 - 4 = 7$	$100(7/20) = 35\%$
$10 \le x < 15$	$17 - 11 = 6$	$100(6/20) = 30\%$
$15 \le x < 20$	$20 - 17 = 3$	$100(3/20) = 15\%$
Total	20	100%

b. Eleven customers certainly waited less than 12 minutes. The six in $[10, 15)$ could all be below 12, all be at or above 12, or be split.

$$\text{Smallest possible count} = 11, \qquad \text{Largest possible count} = 11 + 6 = 17$$

$$\text{Possible percentage range} = \frac{11}{20}(100) \text{ to } \frac{17}{20}(100) = 55\% \text{ to } 85\%$$

$$\text{Count required} = 0.80(20) = 16$$

The target may have been met, but the grouped summary cannot establish that it was. At least five of the six customers in $[10, 15)$ would need to have waited less than 12 minutes. Their individual times, or a count using 12 as a boundary, are needed.

Question 13 (Average)

Nu Retail records one order from each of twelve customers.

Customer	Customer type	Order collection
1	Member	Pickup
2	Member	Pickup
3	Member	Delivery
4	Member	Pickup
5	Member	Delivery
6	Member	Pickup
7	Guest	Pickup
8	Guest	Delivery
9	Guest	Delivery
10	Guest	Delivery
11	Guest	Delivery
12	Guest	Delivery

a. Construct a crosstabulation with customer type in rows and collection method in columns. Include the marginal totals.

b. Construct separate row-percentage and column-percentage tables.

Question 13 (Average)

c. Assess the following statements using your results:

- 80% of pickup customers are members.
- 80% of members use pickup.
- One-third of all recorded customers are members who use pickup.

Question 13: Solution

a. Counts:

Customer type	Pickup	Delivery	Total
Member	4	2	6
Guest	1	5	6
Total	5	7	12

b. Row percentages:

Customer type	Pickup	Delivery	Total
Member	$100(4/6) = 66.67\%$	$100(2/6) = 33.33\%$	100%
Guest	$100(1/6) = 16.67\%$	$100(5/6) = 83.33\%$	100%

Column percentages:

Customer type	Pickup	Delivery
Member	$100(4/5) = 80.00\%$	$100(2/7) = 28.57\%$
Guest	$100(1/5) = 20.00\%$	$100(5/7) = 71.43\%$
Total	100%	100%

Question 13: Solution

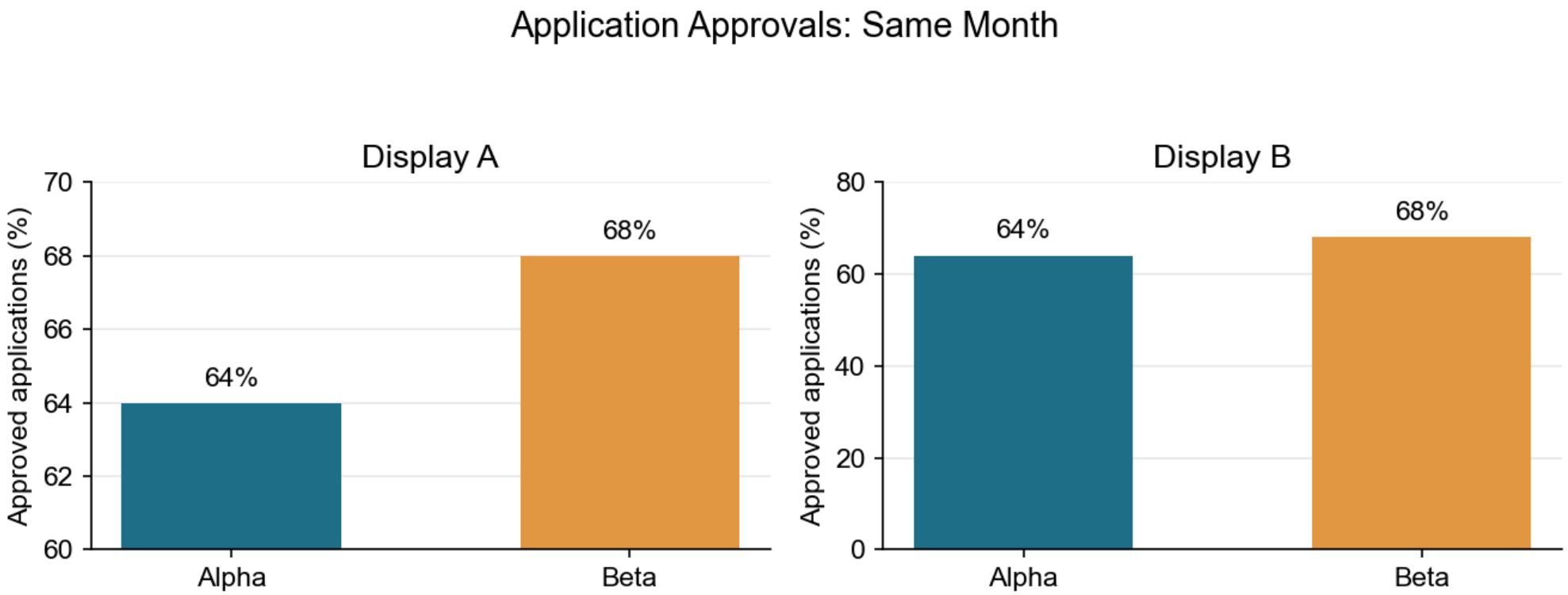
C.

Statement	Calculation	Assessment
Pickup customers who are members	$4/5 = 80\%$	Correct
Members who use pickup	$4/6 = 66.67\%$	Incorrect
All customers who are members using pickup	$4/12 = 1/3$	Correct

The same cell count, 4, answers three different questions when divided by the relevant group total.

Question 14 (Average)

Xi Bank compares application approvals at two branches. Two displays of the same month's records are submitted for its management report.



Alpha: 160 approvals from 250 applications. Beta: 170 approvals from 250 applications.

The proposed commentary reads:

“Beta's greater approval share makes its procedure the better choice. All branches should adopt that procedure next quarter.”

Question 14 (Average)

Which display, if either, would you publish, and what commentary would you attach? Support your choices using the information supplied. No calculations are required.

Question 14: Solution

Publish Display B. Its bars begin at zero, so their lengths provide a proportionate comparison of approval shares. Display A displays the same numerical values but magnifies their visual difference by beginning at 60%.

A defensible commentary is:

“During the reported month, Beta approved 68% of its 250 applications and Alpha approved 64% of its 250 applications.”

The figures do not establish that Beta's procedure is better. The applicant groups may differ, and a greater approval share is not itself evidence of appropriate lending decisions. There is no evidence here about subsequent repayment, costs, or the effect of adopting the procedure elsewhere.

Replacing the chart does not repair the unsupported recommendation. Further evidence about applicant comparability and lending outcomes is needed before a general change.

Question 15 (Hard)

Omicron Maintenance compares two repair teams. Every completed job is classified as routine or complex and as on time or late against its agreed deadline.

Team	Job type	On-time jobs	Completed jobs
Alpha	Routine	18	20
Alpha	Complex	56	80
Beta	Routine	68	80
Beta	Complex	12	20

The operations director recommends Beta for a new contract, citing its overall on-time record.

Evaluate the recommendation numerically. As part of your assessment, compare the teams on a reference workload of 20 routine and 20 complex jobs each, assuming their observed within-type on-time rates apply to that workload. Explain what the records support and what remains uncertain.

Question 15: Solution

On-time rate	Alpha	Beta
Routine	$18/20 = 90\%$	$68/80 = 85\%$
Complex	$56/80 = 70\%$	$12/20 = 60\%$
Overall	$(18 + 56)/(20 + 80) = 74\%$	$(68 + 12)/(80 + 20) = 80\%$

Beta leads overall, but Alpha leads within both job types.

$$\text{Routine-job share: Alpha} = \frac{20}{100} = 20\%, \qquad \text{Beta} = \frac{80}{100} = 80\%$$

Beta's workload contains a much larger share of routine jobs, which have higher on-time rates for both teams. The aggregate ranking reverses the within-type ranking: Simpson's paradox.

For the common reference workload:

Calculation	Alpha	Beta
Routine jobs on time	$20(0.90) = 18$	$20(0.85) = 17$
Complex jobs on time	$20(0.70) = 14$	$20(0.60) = 12$
Total on time	$18 + 14 = 32$	$17 + 12 = 29$
Overall rate	$32/40 = 80.0\%$	$29/40 = 72.5\%$

Alpha has the stronger record for the common workload under the stated assumption. Beta's aggregate rate alone does not justify the recommendation.

Question 15: Solution

These projections are not guaranteed outcomes or proof of a causal team effect. Job differences within each type, capacity, and whether the observed rates will apply to the new contract still need consideration.

Question 16 (Hard)

Pi Stores records 200 customer transactions. Every transaction is classified as promotional or regular and uses exactly one payment method.

The following figures are exact:

Report item	Value
Promotional transactions	120
Card transactions	100
Cash transactions	60
Mobile transactions	40
Promotional share of card transactions	75%
Promotional share of cash transactions	50%

A draft report makes three claims:

1. 75% of promotional transactions use cards.
2. Cards are the most common payment method within each transaction type.
3. There are more regular mobile transactions than promotional mobile transactions.

a. Reconstruct the complete transaction-type by payment-method count table, including its margins.

b. Assess all three claims with calculations. For the mobile comparison, also compare the percentage using mobile within each transaction type.

Question 16: Solution

a.

$$\text{Regular transactions} = 200 - 120 = 80$$

$$\text{Promotional card} = 0.75(100) = 75, \qquad \text{Promotional cash} = 0.50(60) = 30$$

$$\text{Promotional mobile} = 120 - 75 - 30 = 15$$

Subtract the promotional counts from the payment-method totals:

$$\text{Regular card} = 100 - 75 = 25, \quad \text{Regular cash} = 60 - 30 = 30, \quad \text{Regular mobile} = 40 - 15 = 25$$

Transaction type	Card	Cash	Mobile	Total
Promotional	75	30	15	120
Regular	25	30	25	80
Total	100	60	40	200

Question 16: Solution

b.

Claim	Calculation	Assessment
75% of promotional transactions use cards	$100(75/120) = 62.50\%$	Incorrect. The given 75% is $75/100$: the promotional share of card transactions.
Cards are most common within each type	Promotional: $75 > 30 > 15$. Regular: cash $30 >$ card $25 =$ mobile 25.	Incorrect. Cards lead among promotional transactions; cash leads among regular transactions.
There are more regular mobile transactions than promotional mobile transactions	Regular: 25; promotional: 15	Correct

$$\text{Mobile share among regular transactions} = \frac{25}{80}(100) = 31.25\%$$

$$\text{Mobile share among promotional transactions} = \frac{15}{120}(100) = 12.50\%$$

Regular transactions have both the larger mobile count and the larger within-type mobile share.

Chapter 3: Numerical Measures

Questions 17–33.

For percentiles, use $L_p = (p/100)(n+1)$ and interpolate when the location is not an integer. Use sample variance and covariance unless the question identifies a complete population.

Retain unrounded values during calculations. Report final numerical measures to two decimal places when needed and correlation coefficients to three decimal places. State the units.

Question 17 (Easy)

Rho Repairs records the waiting times for eight customers.

Customer	1	2	3	4	5	6	7	8
Waiting time (minutes)	6	9	5	7	6	10	6	15

a. Calculate the mean, median, and mode.

b. A check of the final customer's record changes 15 minutes to 35 minutes. Recalculate the three measures and explain why they do not all respond in the same way.

Question 17: Solution

a. Ordered times: $5, 6, 6, 6, 7, 9, 10, 15$.

$$\bar{x} = \frac{6 + 9 + 5 + 7 + 6 + 10 + 6 + 15}{8} = \frac{64}{8} = 8 \text{ minutes}$$

$$\text{Median} = \frac{6 + 7}{2} = 6.5 \text{ minutes}, \qquad \text{Mode} = 6 \text{ minutes}$$

b. The revised total is $64 - 15 + 35 = 84$.

$$\bar{x}_{new} = \frac{84}{8} = 10.5 \text{ minutes}$$

The middle positions still contain 6 and 7, and 6 remains the most frequent value. The median stays at 6.5 minutes and the mode at 6 minutes.

The mean uses the magnitude of every value. Changing the largest value here does not change the middle positions or which value occurs most often.

Question 18 (Easy)

For a one-period portfolio exercise, use these asset returns as given.

Asset	Return
Alpha	4%
Beta	12%

Divide 10,000 MAD between the two assets at the start of the period. There are no fees or later deposits or withdrawals.

a. Find the investment amounts and portfolio weights that give a portfolio return of 9%.

b. Calculate the portfolio's monetary gain. Does the simple average of the two asset returns give this portfolio's return? Explain.

Question 18: Solution

a. Let w be the initial share invested in Beta. The share in Alpha is $1 - w$.

$$0.04(1 - w) + 0.12w = 0.09$$

$$0.04 + 0.08w = 0.09, \qquad w = \frac{0.05}{0.08} = 0.625$$

$$w_{\text{Alpha}} = 1 - 0.625 = 0.375$$

Asset	Portfolio weight	Initial investment (MAD)
Alpha	37.5%	$0.375(10{,}000) = 3{,}750$
Beta	62.5%	$0.625(10{,}000) = 6{,}250$
Total	100%	10,000

Check the weighted return:

$$r_p = 0.375(0.04) + 0.625(0.12) = 0.015 + 0.075 = 0.09 = 9\%$$

b.

$$\text{Gain from Alpha} = 3{,}750(0.04) = 150 \text{ MAD}$$

$$\text{Gain from Beta} = 6{,}250(0.12) = 750 \text{ MAD}$$

$$\text{Portfolio gain} = 150 + 750 = 900 \text{ MAD}$$

Question 18: Solution

$$\frac{900}{10{,}000} = 0.09 = 9\%$$

The simple average is

$$\frac{4\% + 12\%}{2} = 8\%.$$

That return corresponds to equal investment weights, not this portfolio's 37.5% and 62.5%. The construction uses the given returns; it does not establish a guaranteed future return.

Question 19 (Easy)

Tau Couriers records processing times for nine parcels.

Parcel	1	2	3	4	5	6	7	8	9
Processing time (minutes)	22	8	14	30	10	18	12	26	16

Calculate P_{40} and P_{65}. Show the ordered data, each location, and any interpolation. Explain what a non-integer location represents.

Question 19: Solution

Ordered position	1	2	3	4	5	6	7	8	9
Time (minutes)	8	10	12	14	16	18	22	26	30

$$L_{40} = 0.40(9+1) = 4, \qquad P_{40} = x_{(4)} = 14 \text{ minutes}$$

$$L_{65} = 0.65(9+1) = 6.5$$

$$P_{65} = x_{(6)} + 0.5[x_{(7)} - x_{(6)}] = 18 + 0.5(22 - 18) = 20 \text{ minutes}$$

Location 6.5 lies halfway between the sixth and seventh ordered observations. It is a position, not a processing time.

Question 20 (Easy)

Upsilon Engineering records the following downtime for all four machines in one workshop during a month.

Machine	Alpha	Beta	Gamma	Delta
Downtime (hours)	2	4	4	6

a. Treat these four machines as the complete population of interest. Calculate the mean, population variance, and population standard deviation.

b. Instead, treat the four machines as a sample used to estimate variability across the company's larger machine fleet. Calculate the sample variance and sample standard deviation. Explain why the numerical results change although the observations do not.

Question 20: Solution

a.

$$\mu = \frac{2+4+4+6}{4} = 4 \text{ hours}$$

$$\sum (x_i - \mu)^2 = (2-4)^2 + (4-4)^2 + (4-4)^2 + (6-4)^2 = 8 \text{ hours}^2$$

$$\sigma^2 = \frac{8}{4} = 2 \text{ hours}^2, \qquad \sigma = \sqrt{2} \approx 1.41 \text{ hours}$$

b. The sample mean is also 4 hours and the squared-deviation total is unchanged.

$$s^2 = \frac{8}{4-1} = \frac{8}{3} \approx 2.67 \text{ hours}^2$$

$$s = \sqrt{\frac{8}{3}} \approx 1.63 \text{ hours}$$

The population calculation describes the four machines themselves. The sample calculation uses $n - 1$ when estimating population variability from those observations. This denominator change does not establish that the workshop represents the whole fleet.

Question 21 (Easy)

Phi Retail compares a day's sales at two branches with each branch's previous daily records.

Branch	Sales on the day (thousand MAD)	Previous sample mean	Previous sample standard deviation
Alpha	220	200	20
Beta	90	80	5

a. Calculate the day's z-score for each branch. Which branch has the higher sales, and which is further above its own usual level?

b. What sales figure would Alpha need to have the same z-score as Beta on this day?

Question 21: Solution

a.

$$z_{\text{Alpha}} = \frac{220 - 200}{20} = 1, \qquad z_{\text{Beta}} = \frac{90 - 80}{5} = 2$$

Alpha has higher sales in thousand MAD. Beta is further above its own mean in standard-deviation units. The z-scores are unitless.

b.

$$\frac{x - 200}{20} = 2$$

$$x - 200 = 2(20) = 40, \qquad x = 240 \text{ thousand MAD}$$

The comparison concerns relative position, not profit or the cause of the sales difference.

Question 22 (Average)

Chi Investments reports three successive annual returns. No money is added or withdrawn.

Year	1	2	3
Return	20%	-10%	5%

a. Calculate the arithmetic mean annual return and the geometric mean annual return.

b. Starting with 1,000 MAD, calculate the ending value after the three actual returns. Which average reproduces this ending value when applied as a constant annual rate?

c. Quantify the difference in ending value if the other average is used as a constant rate.

Question 22: Solution

a.

$$\bar{r} = \frac{20 - 10 + 5}{3} = 5\%$$

$$\prod(1 + r_t) = 1.20(0.90)(1.05) = 1.134$$

$$\bar{r}_G = (1.134)^{1/3} - 1 \approx 0.042808 = 4.28\%$$

b.

Point in time	Value (MAD)
Initial	1,000
End of Year 1	$1{,}000(1.20) = 1{,}200$
End of Year 2	$1{,}200(0.90) = 1{,}080$
End of Year 3	$1{,}080(1.05) = 1{,}134$

Using the unrounded geometric mean:

$$1{,}000(1 + \bar{r}_G)^3 = 1{,}000(1.134) = 1{,}134 \text{ MAD}$$

c.

$$1{,}000(1.05)^3 = 1{,}157.625 \text{ MAD}$$

Question 22: Solution

$$\text{Overstatement} = 1{,}157.625 - 1{,}134 = 23.625 \approx 23.63 \text{ MAD}$$

The arithmetic mean averages the annual returns. It does not reproduce their compounded growth.

Question 23 (Average)

Psi Distribution records delivery costs for ten shipments.

Shipment	1	2	3	4	5	6	7	8	9	10
Cost (MAD)	24	13	31	19	11	26	39	18	22	17

a. Calculate Q_1, Q_2, Q_3, and the IQR using the lecture's percentile convention.

b. An analyst states that exactly 25% of the recorded costs must be below the calculated Q_1. Assess the statement using these observations.

Question 23: Solution

a.

Ordered position	1	2	3	4	5	6	7	8	9	10
Cost (MAD)	11	13	17	18	19	22	24	26	31	39

$$L_{25} = 0.25(11) = 2.75, \qquad Q_1 = 13 + 0.75(17 - 13) = 16 \text{ MAD}$$

$$L_{50} = 0.50(11) = 5.5, \qquad Q_2 = 19 + 0.5(22 - 19) = 20.50 \text{ MAD}$$

$$L_{75} = 0.75(11) = 8.25, \qquad Q_3 = 26 + 0.25(31 - 26) = 27.25 \text{ MAD}$$

$$IQR = 27.25 - 16 = 11.25 \text{ MAD}$$

b. Only 11 and 13 are below 16.

$$\text{Observed share below } Q_1 = \frac{2}{10}(100) = 20\%$$

The statement is incorrect. An interpolated percentile identifies a position in the ordered distribution; it need not leave exactly that percentage of a small sample below it.

Question 24 (Average)

Omega Foods uses two packaging lines for different bag sizes.

Line	Sample mean net weight (g)	Sample standard deviation (g)
Alpha	40	2
Beta	80	3

The quality manager describes Alpha as more consistent. The production manager describes Beta as more consistent.

a. Calculate each line's coefficient of variation.

b. Assess both descriptions, specifying the meaning of consistency under which each could be defended.

c. Can these summaries identify which line produces a larger percentage of bags outside its permitted weight limits? Explain.

Question 24: Solution

a.

$$CV_{\text{Alpha}} = \frac{2}{40}(100) = 5\%, \qquad CV_{\text{Beta}} = \frac{3}{80}(100) = 3.75\%$$

b. Alpha has the smaller absolute standard deviation: 2 g rather than 3 g. Beta has the smaller standard deviation relative to its mean: 3.75% rather than 5%.

Both descriptions can be defended, but they use different measures of consistency. Net weight has a meaningful zero and the means are positive, so CV is appropriate for this relative comparison.

c. No. The permitted limits and the distributions or individual weights are not supplied. Mean and standard deviation alone do not determine the percentage outside unspecified limits.

Question 25 (Average)

Alpha Dispatch records five delivery times for each of two routes.

Delivery	1	2	3	4	5
Route A (minutes)	8	9	10	11	12
Route B (minutes)	6	8	10	12	14

The contract requires delivery within 12 minutes, including delivery at exactly 12 minutes.

a. Calculate the sample mean, range, variance, and standard deviation for each route.

b. Calculate the average absolute distance from the mean for each route. Is that quantity the same as the standard deviation?

c. Compare the recorded proportions meeting the contract. Assess the claim that equal mean delivery times imply equivalent service.

Question 25: Solution

a.

$$\bar{x}_A = \frac{8+9+10+11+12}{5} = 10, \qquad \bar{x}_B = \frac{6+8+10+12+14}{5} = 10 \text{ minutes}$$

$$Range_A = 12 - 8 = 4, \qquad Range_B = 14 - 6 = 8 \text{ minutes}$$

$$s_A^2 = \frac{(-2)^2 + (-1)^2 + 0^2 + 1^2 + 2^2}{4} = \frac{10}{4} = 2.50 \text{ minutes}^2$$

$$s_B^2 = \frac{(-4)^2 + (-2)^2 + 0^2 + 2^2 + 4^2}{4} = \frac{40}{4} = 10 \text{ minutes}^2$$

$$s_A = \sqrt{2.5} \approx 1.58, \qquad s_B = \sqrt{10} \approx 3.16 \text{ minutes}$$

b.

$$\text{Average absolute distance, A} = \frac{2+1+0+1+2}{5} = 1.20 \text{ minutes}$$

$$\text{Average absolute distance, B} = \frac{4+2+0+2+4}{5} = 2.40 \text{ minutes}$$

These are not the standard deviations. Standard deviation uses squared deviations, with the sample variance divided by $n - 1$, followed by a square root.

Question 25: Solution

c.

$$\text{Within 12 minutes, A} = \frac{5}{5} = 100\%, \qquad \text{Within 12 minutes, B} = \frac{4}{5} = 80\%$$

The means are equal, but Route B is more variable and has a lower observed compliance rate. These five deliveries do not establish the long-run rates.

Question 26 (Average)

Beta Components examines a complete batch of 10,000 items. Their processing times have population mean 50 seconds and population standard deviation 4 seconds.

a. Without assuming a distribution shape, what minimum percentage and minimum number of items can be guaranteed to have processing times in the inclusive interval 40–60 seconds?

b. Suppose a histogram of the batch is approximately bell-shaped. Estimate the number in the inclusive interval 42–58 seconds using the empirical rule.

c. A report changes the estimate in part b to “at least 9,500 items.” Assess this wording and calculate the minimum count Chebyshev's theorem guarantees for the same interval.

Question 26: Solution

a. The interval extends 10 seconds on either side of the mean.

$$k = \frac{10}{4} = 2.5$$

$$\text{Minimum proportion} = 1 - \frac{1}{2.5^2} = 1 - \frac{1}{6.25} = 0.84$$

$$\text{Minimum percentage} = 84\%, \qquad \text{Minimum count} = 0.84(10{,}000) = 8{,}400$$

b.

$$k = \frac{8}{4} = 2$$

For an approximately bell-shaped distribution, approximately 95% lie within two standard deviations:

$$\text{Estimated count} = 0.95(10{,}000) = 9{,}500$$

c. “At least” is not justified by the empirical rule. Its 95% is an approximation under the shape assumption, not a guaranteed lower bound.

$$\text{Chebyshev minimum} = 1 - \frac{1}{2^2} = 0.75$$

$$\text{Guaranteed minimum count} = 0.75(10{,}000) = 7{,}500$$

Question 27 (Average)

Gamma Service records the following waiting times for nine customers.

Customer	1	2	3	4	5	6	7	8	9
Waiting time (minutes)	13	8	30	11	6	16	10	14	12

Calculate the five-number summary and the IQR fences. Sketch a box plot with potential outliers shown separately. State the two observed values at which the lines extending from the box end.

Question 27: Solution

Ordered times: $6, 8, 10, 11, 12, 13, 14, 16, 30$.

$$L_{25} = 0.25(10) = 2.5, \qquad Q_1 = 8 + 0.5(10 - 8) = 9$$

$$L_{50} = 0.50(10) = 5, \qquad Q_2 = 12$$

$$L_{75} = 0.75(10) = 7.5, \qquad Q_3 = 14 + 0.5(16 - 14) = 15$$

Minimum	Q_1	Median	Q_3	Maximum
6	9	12	15	30

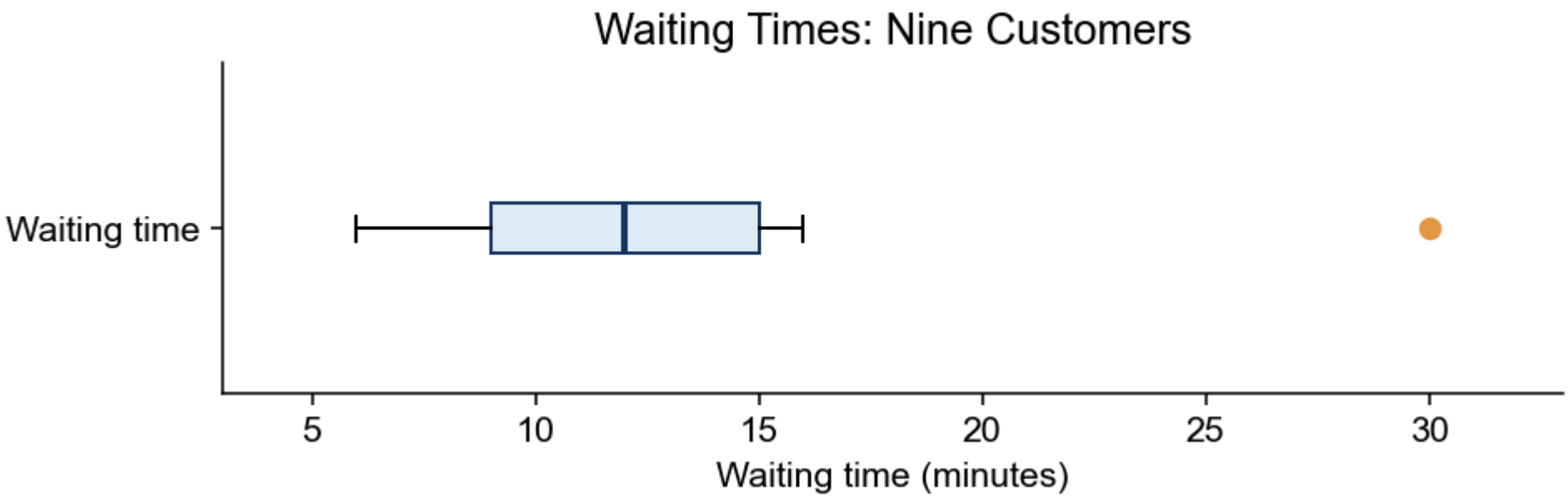
All values are in minutes.

$$IQR = 15 - 9 = 6$$

$$\text{Lower fence} = 9 - 1.5(6) = 0, \qquad \text{Upper fence} = 15 + 1.5(6) = 24$$

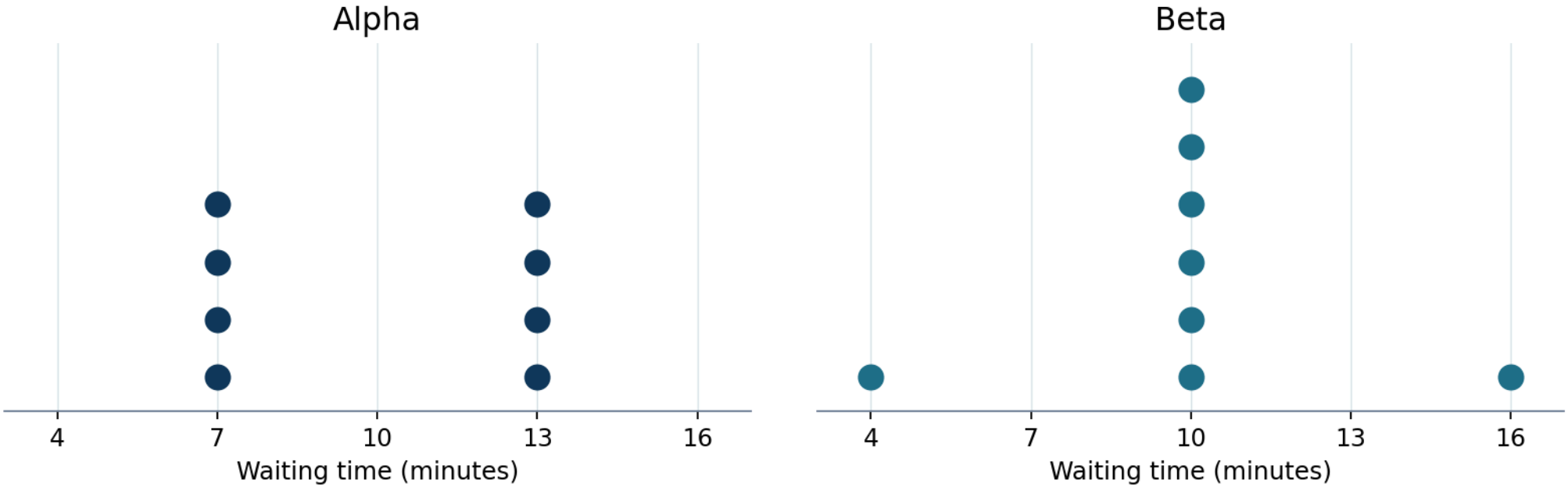
The value 30 is above 24 and is plotted separately. The lines end at 6 and 16, the smallest and largest observed values within the fences. The upper endpoint is not 24 because 24 is a cutoff, not an observed waiting time.

Question 27: Solution



Question 28 (Average)

Delta Support compares eight recorded waiting times for each of two teams.



Each dot represents one observation.

Measure	Alpha	Beta
Mean	10 minutes	10 minutes
Median	10 minutes	10 minutes
Sample standard deviation	Approximately 3.21 minutes	Approximately 3.21 minutes

Question 28 (Average)

Compare the waiting-time patterns shown in the plots. Explain what the matching summaries conceal and whether a bell-shaped description is appropriate for either set of observations. No calculations are required.

Question 28: Solution

Alpha has two equal clusters at 7 and 13 minutes, with no observations near the mean. Beta has six observations at 10 minutes and two isolated values at 4 and 16 minutes.

The matching mean and median locate the centre. The matching standard deviations describe equal variability by that measure, but do not show where observations are concentrated, the gaps between them, or how far the most extreme values extend.

The plots do not justify a bell-shaped description: Alpha has two separated clusters, while Beta has a central stack and two isolated extremes. Equal mean and median do not establish a bell shape, and these small samples do not establish the population shapes.

Question 29 (Average)

Epsilon Manufacturing records machine age and average output for five machines under the observed operating conditions.

Machine	Age, X (years)	Output, Y (units per minute)
Alpha	1	8
Beta	2	6
Gamma	3	7
Delta	4	3
Epsilon	5	1

Calculate the sample covariance and correlation. Show the deviations and paired cross-products in a calculation table.

State the association supported by these records. Do the results alone establish how much output would improve if an old machine were replaced?

Question 29: Solution

$$\bar{x} = \frac{1+2+3+4+5}{5} = 3 \text{ years}, \qquad \bar{y} = \frac{8+6+7+3+1}{5} = 5 \text{ units per minute}$$

X	Y	$X-3$	$Y-5$	$(X-3)^2$	$(Y-5)^2$	$(X-3)(Y-5)$
1	8	-2	3	4	9	-6
2	6	-1	1	1	1	-1
3	7	0	2	0	4	0
4	3	1	-2	1	4	-2
5	1	2	-4	4	16	-8
Total		0	0	10	34	-17

$$s_{XY} = \frac{-17}{5-1} = -4.25 \text{ years} \times \text{units per minute}$$

$$s_X = \sqrt{\frac{10}{4}} = \sqrt{2.5}, \qquad s_Y = \sqrt{\frac{34}{4}} = \sqrt{8.5}$$

$$r = \frac{s_{XY}}{s_X s_Y} = \frac{-4.25}{\sqrt{2.5}\sqrt{8.5}} \approx -0.922$$

The records show a strong negative linear association. They do not isolate the effect of age from machine type, maintenance, workload, or other conditions, so they do not establish the improvement from replacement.

Question 30 (Average)

Zeta Services reports a mean job cost of 25 MAD from eight jobs. A check finds that one cost was entered as 48 MAD rather than its correct value of 24 MAD. The other seven costs were recorded correctly.

A ninth job, costing 54 MAD, is then added.

a. Calculate the corrected mean for the original eight jobs and the mean after adding the ninth.

b. Can the median of the nine correct costs be determined from this information? Justify your answer.

Question 30: Solution

a.

$$\text{Original recorded total} = 8(25) = 200 \text{ MAD}$$

$$\text{Corrected total} = 200 - 48 + 24 = 176 \text{ MAD}$$

$$\bar{x}_8 = \frac{176}{8} = 22 \text{ MAD}$$

$$\bar{x}_9 = \frac{176 + 54}{9} = \frac{230}{9} \approx 25.56 \text{ MAD}$$

b. No. The total does not determine the ordered positions. For example, either of these could be the eight corrected costs:

Possible dataset	Eight costs in ascending order	Total
A	10, 14, 18, 20, 22, 24, 30, 38	176
B	8, 12, 16, 24, 26, 28, 30, 32	176

Both contain the corrected value 24, and replacing it with 48 would produce the reported total of 200.

After adding 54, the fifth ordered value is 22 for A but 26 for B. Thus the same supplied information allows different medians.

Question 31 (Hard)

Eta Logistics records six loading delays.

Observation	1	2	3	4	5	6
Delay, X (minutes)	1	1	2	2	3	9

a. Calculate the mean, direct moment skewness, Pearson kurtosis, and excess kurtosis. Show the deviations and their second, third, and fourth powers in a calculation table.

b. The company assigns each observation a performance score $Y = 10 - X$, where X is the delay measured in minutes and higher scores are preferred. Determine the score's skewness, Pearson kurtosis, and excess kurtosis from the relationship between X and Y. Justify your results without constructing another full deviation table.

c. After switching to the score, a manager reports that the distribution's shape demonstrates an improvement in operations. Assess the claim and interpret what the calculated measures establish about these observations.

Question 31: Solution

a.

$$\bar{x} = \frac{1+1+2+2+3+9}{6} = \frac{18}{6} = 3 \text{ minutes}$$

Observation	x_i	$x_i - 3$	$(x_i - 3)^2$	$(x_i - 3)^3$	$(x_i - 3)^4$
1	1	-2	4	-8	16
2	1	-2	4	-8	16
3	2	-1	1	-1	1
4	2	-1	1	-1	1
5	3	0	0	0	0
6	9	6	36	216	1,296
Total	18	0	46	198	1,330

For direct moments, $m_r = \frac{1}{n}\sum(x_i - \bar{x})^r$.

$$m_2 = \frac{46}{6} = \frac{23}{3}, \qquad m_3 = \frac{198}{6} = 33, \qquad m_4 = \frac{1{,}330}{6} = \frac{665}{3}$$

$$g_1 = \frac{m_3}{m_2^{3/2}} = \frac{33}{(23/3)^{3/2}} \approx 1.55$$

Question 31: Solution

$$b_2 = \frac{m_4}{m_2^2} = \frac{665/3}{(23/3)^2} = \frac{1{,}995}{529} \approx 3.77$$

$$g_2 = b_2 - 3 = \frac{408}{529} \approx 0.77$$

b. The mean score is

$$\bar{y} = \frac{\sum(10 - x_i)}{6} = \frac{60 - 18}{6} = 7.$$

For each observation,

$$y_i - \bar{y} = (10 - x_i) - (10 - \bar{x}) = -(x_i - \bar{x}).$$

Squaring or taking the fourth power removes the minus sign; cubing preserves it. Therefore,

$$m_{2,Y} = m_{2,X} = \frac{23}{3}, \qquad m_{3,Y} = -m_{3,X} = -33, \qquad m_{4,Y} = m_{4,X} = \frac{665}{3}.$$

$$g_{1,Y} = \frac{-33}{(23/3)^{3/2}} = -g_{1,X} \approx -1.55$$

$$b_{2,Y} = \frac{665/3}{(23/3)^2} = b_{2,X} \approx 3.77$$

$$g_{2,Y} = b_{2,Y} - 3 \approx 0.77$$

Question 31: Solution

c. The delays are right-skewed; the scores are left-skewed. The nine-minute delay becomes a score of $10 - 9 = 1$, so the long delay in the right tail becomes a low score in the left tail.

No loading delay has changed. Reversing the scale changes the sign of skewness, not the underlying performance. Both representations have the same Pearson and excess kurtosis, indicating greater tail extremity relative to the normal reference.

These six observations describe the recorded delays. They do not establish an improvement over an earlier period or a particular population distribution.

Question 32 (Hard)

Theta Repairs records waiting times for a sample of ten jobs.

Job	1	2	3	4	5	6	7	8	9	10
Waiting time (minutes)	12	9	38	11	14	10	13	12	10	11

One analyst says that 38 minutes is not unusual because it is within three sample standard deviations of the mean. Another wants it deleted from the service report as an error. The original job log has not yet been checked.

Assess both positions numerically using the z-score and IQR criteria. Quantify how retaining or omitting 38 changes the mean and sample standard deviation, and recommend what should be reported while the record is investigated.

Question 32: Solution

For all ten observations:

$$\bar{x} = \frac{140}{10} = 14 \text{ minutes}$$

$$\sum (x_i - 14)^2 = 25 + 16 + 16 + 9 + 9 + 4 + 4 + 1 + 0 + 576 = 660$$

$$s^2 = \frac{660}{9}, \qquad s = \sqrt{\frac{660}{9}} \approx 8.56 \text{ minutes}$$

$$z(38) = \frac{38 - 14}{\sqrt{660/9}} \approx 2.80$$

The value is not flagged by the $|z| > 3$ criterion.

Ordered times: $9, 10, 10, 11, 11, 12, 12, 13, 14, 38$.

$$Q_1 = 10 + 0.75(10 - 10) = 10, \qquad Q_3 = 13 + 0.25(14 - 13) = 13.25$$

$$IQR = 13.25 - 10 = 3.25$$

$$\text{Lower fence} = 10 - 1.5(3.25) = 5.125$$

$$\text{Upper fence} = 13.25 + 1.5(3.25) = 18.125$$

The IQR criterion flags 38. Failure to exceed three sample standard deviations does not establish that the observation is ordinary under every criterion.

Question 32: Solution

If 38 is omitted:

$$n = 9, \qquad \bar{x}_{omit} = \frac{140 - 38}{9} = \frac{102}{9} = \frac{34}{3} \approx 11.33 \text{ minutes}$$

The remaining values are $9, 10, 10, 11, 11, 12, 12, 13, 14$:

$$\sum (x_i - 34/3)^2 = \frac{49 + 16 + 16 + 1 + 1 + 4 + 4 + 25 + 64}{9} = 20$$

$$s_{omit} = \sqrt{\frac{20}{8}} = \sqrt{2.5} \approx 1.58 \text{ minutes}$$

Treatment	Mean (minutes)	Sample standard deviation (minutes)
Retain 38	14.00	8.56
Omit 38	11.33	1.58

Including 38 raises the mean from 11.33 to 14 minutes and the standard deviation from 1.58 to 8.56 minutes. It therefore moves the mean closer to itself and enlarges the standard deviation used to calculate its own z-score. Both effects reduce that z-score, explaining why the three-standard-deviation check does not flag it.

Neither criterion establishes a recording error. Keep the observation in the main report pending verification, flag it for investigation, and label the omission results as a sensitivity comparison. Correct a verified error; retain a genuine delay belonging to the population being described.

Question 33 (Hard)

Iota Retail records advertising expenditure and orders for five stores during the same month.

Store	Advertising, X (thousand MAD)	Orders, Y
Alpha	2	40
Beta	4	120
Gamma	6	80
Delta	8	100
Epsilon	10	60

Before calculating correlation, an analyst sorts each numerical column separately from smallest to largest. The report then states:

“Advertising and orders have a perfect positive correlation. Sorting cannot change the result because it leaves each variable's mean and standard deviation unchanged.”

a. Audit this statement. Calculate the sample covariance and correlation from the original pairs and from the separately sorted columns. Show the paired calculations.

b. Using the original pairs, express advertising expenditure in MAD rather than thousand MAD. Determine the resulting covariance and correlation without treating the unit change as new observations.

c. State the conclusion about advertising and orders that the original records support.

Question 33: Solution

a. Original pairs:

$$\bar{x} = \frac{2+4+6+8+10}{5} = 6 \text{ thousand MAD}, \qquad \bar{y} = \frac{40+120+80+100+60}{5} = 80 \text{ orders}$$

X	Y	$X-6$	$Y-80$	$(X-6)^2$	$(Y-80)^2$	$(X-6)(Y-80)$
2	40	-4	-40	16	1,600	160
4	120	-2	40	4	1,600	-80
6	80	0	0	0	0	0
8	100	2	20	4	400	40
10	60	4	-20	16	400	-80
Total		0	0	40	4,000	40

$$s_{XY} = \frac{40}{4} = 10 \text{ (thousand MAD)} \times \text{orders}$$

$$s_X = \sqrt{\frac{40}{4}} = \sqrt{10}, \qquad s_Y = \sqrt{\frac{4{,}000}{4}} = \sqrt{1{,}000}$$

$$r = \frac{10}{\sqrt{10}\sqrt{1{,}000}} = 0.100$$

Question 33: Solution

After sorting the columns separately:

X	Reordered Y	$X-6$	$Y-80$	$(X-6)(Y-80)$
2	40	-4	-40	160
4	60	-2	-20	40
6	80	0	0	0
8	100	2	20	40
10	120	4	40	160
Total		0	0	400

The separate means and squared-deviation totals are unchanged, but the cross-products change:

$$s_{XY,\,sorted} = \frac{400}{4} = 100, \qquad r_{sorted} = \frac{100}{\sqrt{10}\sqrt{1{,}000}} = 1.000$$

The analyst's calculation describes artificial pairings, not the five stores. Sorting complete rows together would preserve the pairings; sorting the two columns independently does not.

b. Let $X' = 1{,}000X$, so $\bar{x}' = 1{,}000(6) = 6{,}000$ MAD and each advertising deviation is multiplied by 1,000.

$$\sum (X' - \bar{x}')(Y - \bar{y}) = 1{,}000(40) = 40{,}000$$

Question 33: Solution

$$s_{X'Y} = \frac{40{,}000}{4} = 10{,}000 \text{ MAD} \times \text{orders}$$

$$s_{X'} = \sqrt{\frac{1{,}000^2(40)}{4}} = 1{,}000\sqrt{10}$$

$$r_{X'Y} = \frac{10{,}000}{(1{,}000\sqrt{10})\sqrt{1{,}000}} = 0.100$$

c. The original records show a weak positive linear association, not a perfect relationship. They do not establish that advertising caused the order differences or that changing advertising would produce a particular increase in orders.

Chapter 4: Introduction to Probability

Questions 34–50.

State the events used in probability calculations. Give exact fractions where convenient; otherwise report probabilities to four decimal places and monetary amounts to two decimal places when needed. Retain unrounded values during calculations.

Do not assume equally likely outcomes or independence unless the information supports those assumptions.

Question 34 (Easy)

Kappa Sales contacts two different customers in a specified order. Record purchase (P) or no purchase (N) for each customer. The four outcomes below exhaust the possibilities.

Outcome, first customer then second	Assigned probability
(P, P)	0.10
(P, N)	0.20
(N, P)	0.25
(N, N)	p

a. Find p and verify that the assignments form a valid probability model.

b. List the outcomes in each event and calculate its probability: the first customer purchases; exactly one customer purchases; at least one customer purchases.

c. An assistant assigns probability $1/4$ to each outcome because there are four rows. Assess that assignment.

Question 34: Solution

a.

$$p = 1 - (0.10 + 0.20 + 0.25) = 0.45$$

All four probabilities are nonnegative, no probability exceeds 1, and their sum is 1.

b.

Event	Outcomes	Probability
First customer purchases	$(P, P), (P, N)$	$0.10 + 0.20 = 0.30$
Exactly one purchases	$(P, N), (N, P)$	$0.20 + 0.25 = 0.45$
At least one purchases	$(P, P), (P, N), (N, P)$	$0.10 + 0.20 + 0.25 = 0.55$

Check the last result by the complement rule:

$$P(\text{at least one purchase}) = 1 - P(N, N) = 1 - 0.45 = 0.55$$

c. Four possible outcomes do not establish equal likelihood. Assigning $1/4$ to every row contradicts the supplied probabilities.

Question 35 (Easy)

Lambda Delivery sells the following service configurations.

Service	Available dispatch days	Available parcel sizes
Economy	Monday, Wednesday, Friday	Small, large
Priority	Tuesday, Thursday	Small, medium, large

For either service, the customer also chooses prepaid or invoice billing. Every combination within a service is available. A configuration specifies the service, dispatch day, parcel size, and billing method.

a. How many configurations are available altogether?

b. How many use invoice billing and a large parcel?

c. A catalogue designer multiplies 2 services by 5 days by 3 sizes by 2 billing methods and reports 60 configurations. Explain the error.

Question 35: Solution

a. Count within each service before adding the nonoverlapping groups.

$$N_{\text{Economy}} = 3(2)(2) = 12$$

$$N_{\text{Priority}} = 2(3)(2) = 12$$

$$N_{\text{total}} = 12 + 12 = 24$$

b. Fix large size and invoice billing:

$$N = 3(1)(1) + 2(1)(1) = 5$$

c. The 60-count calculation treats every day and size as available with both services. For example, Economy on Tuesday and Economy with a medium parcel are not offered. The available choices depend on the selected service.

Question 36 (Easy)

Mu Consulting has six distinct employees. Four, including Alpha, are certified to lead an assignment; two are not. Any employee can act as reviewer.

Each assignment needs one lead and one reviewer, and the same employee cannot hold both roles.

a. How many lead–reviewer assignments would be possible without the certification requirement?

b. How many meet the certification requirement?

c. How many valid assignments give Alpha one of the two roles?

Question 36: Solution

a. Interchanging the lead and reviewer gives a different assignment.

$$P(6, 2) = 6(5) = 30$$

b. Choose a certified lead, then a different reviewer:

$$4(5) = 20$$

c. Separate Alpha's two possible roles.

Alpha's role	Choices for the other role	Assignments
Lead	Any of the other five employees as reviewer	5
Reviewer	One of the other three certified employees as lead	3

Alpha cannot hold both roles, so the cases do not overlap.

$$N = 5 + 3 = 8$$

Question 37 (Easy)

Nu Investments will fund three distinct projects from the following pool.

Project category	Number available
Energy	3
Logistics	4
Total	7

Every three-project group is affordable and equally likely to be selected. A project cannot be selected twice, and the order of selection is ignored.

a. How many groups are possible?

b. How many contain exactly two energy projects? What is the probability of selecting such a group?

c. Calculate the probability that the selected group contains at least one energy project.

Question 37: Solution

a.

$$N = \binom{7}{3} = \frac{7(6)(5)}{3(2)(1)} = 35$$

b. Select two of the three energy projects and one of the four logistics projects.

$$N_{\text{two energy}} = \binom{3}{2}\binom{4}{1} = 3(4) = 12$$

$$P(\text{exactly two energy}) = \frac{12}{35} \approx 0.3429$$

c. A group with no energy projects contains three logistics projects.

$$N_{\text{no energy}} = \binom{4}{3} = 4$$

$$P(\text{at least one energy}) = 1 - \frac{4}{35} = \frac{31}{35} \approx 0.8857$$

Question 38 (Easy)

Xi Installations reviews 80 completed contracts.

Completion	Rework required	No rework	Total
Late	6	14	20
On time	10	50	60
Total	16	64	80

Select one of these contracts uniformly at random. Let L denote late completion and R denote rework.

Calculate the probability that the contract:

a. Was late.

b. Was late and required rework.

c. Had at least one of the two problems.

d. Had exactly one of the two problems.

e. Had neither problem.

Question 38: Solution

a.

$$P(L) = \frac{20}{80} = 0.25$$

b.

$$P(L \cap R) = \frac{6}{80} = 0.075$$

c.

$$P(L \cup R) = \frac{20 + 16 - 6}{80} = \frac{30}{80} = 0.375$$

d. Exactly one problem excludes the six contracts with both.

$$P(\text{exactly one}) = \frac{14 + 10}{80} = \frac{24}{80} = 0.30$$

e.

$$P(L^c \cap R^c) = \frac{50}{80} = 0.625$$

Check:

$$P(L \cup R) + P(L^c \cap R^c) = 0.375 + 0.625 = 1$$

Question 39 (Average)

Omicron Claims has eight distinct claim files: three flagged and five unflagged. Two files are selected sequentially, uniformly at each draw, without replacement.

a. Calculate the probability that both selected files are flagged.

b. Calculate the probability that exactly one selected file is flagged.

c. Given that at least one selected file is flagged, calculate the probability that exactly one is flagged.

d. If the first file is replaced before the second draw, what is the probability that both selections are flagged? Compare it with part a.

Question 39: Solution

Let F_1 and F_2 denote flagged files on the first and second draws.

For parts (a)–(c), sampling is without replacement:

First draw	Second draw	Event	Probability
Flagged	Flagged	$F_1 \cap F_2$	$\frac{3}{8}\frac{2}{7} = \frac{6}{56}$
Flagged	Unflagged	$F_1 \cap F_2^c$	$\frac{3}{8}\frac{5}{7} = \frac{15}{56}$
Unflagged	Flagged	$F_1^c \cap F_2$	$\frac{5}{8}\frac{3}{7} = \frac{15}{56}$
Unflagged	Unflagged	$F_1^c \cap F_2^c$	$\frac{5}{8}\frac{4}{7} = \frac{20}{56}$
Total			1

a. After a flagged first draw, two flagged files remain among seven files.

$$P(F_1 \cap F_2) = P(F_1)P(F_2 \mid F_1) = \frac{3}{8}\frac{2}{7} = \frac{3}{28} \approx 0.1071$$

b. Exactly one flag corresponds to the second and third rows.

$$P(\text{exactly one}) = \frac{3}{8}\frac{5}{7} + \frac{5}{8}\frac{3}{7} = \frac{30}{56} = \frac{15}{28} \approx 0.5357$$

c. At least one flag covers the first three outcome rows; its complement is two unflagged files.

Question 39: Solution

$$P(\text{at least one}) = 1 - \frac{5}{8}\frac{4}{7} = 1 - \frac{20}{56} = \frac{18}{28}$$

Exactly one flagged file is a subset of at least one flagged file.

$$P(\text{exactly one} \mid \text{at least one}) = \frac{15/28}{18/28} = \frac{5}{6} \approx 0.8333$$

d. With replacement, the second draw again has three flagged files out of eight.

$$P(\text{both flagged}) = \frac{3}{8}\frac{3}{8} = \frac{9}{64} \approx 0.1406$$

This exceeds $3/28$. Without replacement, a flagged first draw reduces the flagged share available for the second draw.

Question 40 (Average)

Pi Freight records the following 80 shipments.

Destination	Insured	Not insured	Total
Domestic	30	18	48
International	20	12	32
Total	50	30	80

Select one recorded shipment uniformly at random. Let I denote international and C denote insured.

a. Calculate $P(I \mid C)$ and $P(C \mid I)$ and interpret each.

b. Determine whether I and C are independent in these records. Support the answer numerically.

c. Are the events mutually exclusive? Explain how that answer can be consistent with part b.

Question 40: Solution

a.

$$P(I \mid C) = \frac{20}{50} = 0.40$$

Forty percent of insured shipments are international.

$$P(C \mid I) = \frac{20}{32} = 0.625$$

Among international shipments, 62.5% are insured. Reversing the condition changes the group being described.

b.

$$P(I) = \frac{32}{80} = 0.40, \qquad P(C) = \frac{50}{80} = 0.625$$

$$P(I \cap C) = \frac{20}{80} = 0.25$$

$$P(I)P(C) = 0.40(0.625) = 0.25 = P(I \cap C)$$

The events are independent for a random selection from these records. Also, $P(C \mid I) = P(C)$.

c. They are not mutually exclusive: 20 shipments are both international and insured. Independence permits overlap; here the overlap has exactly the probability given by the product of the marginal probabilities.

Question 41 (Average)

Rho Retail receives a probability report about individual orders.

Event	Reported probability
A: customer uses the app	0.60
B: customer uses a discount	0.55
Both A and B	0.10

a. Can all three probabilities describe the same order population? Justify your answer numerically.

b. Keep $P(A) = 0.60$ and $P(B) = 0.55$ unchanged. Determine the full range of possible values for $P(A \cap B)$.

c. If A and B were independent, calculate the probability of both, neither, and exactly one event.

Question 41: Solution

a. The reported overlap gives

$$P(A \cup B) = 0.60 + 0.55 - 0.10 = 1.05$$

A probability cannot exceed 1. Equivalently, the implied probability of neither is $1 - 1.05 = -0.05$. The report is inconsistent.

b. Write $x = P(A \cap B)$.

Outcome	Probability
Both	x
App only	$0.60 - x$
Discount only	$0.55 - x$
Neither	$1 - 0.60 - 0.55 + x = x - 0.15$

Nonnegative cell probabilities require

$$x \geq 0.15, \qquad x \leq 0.60, \qquad x \leq 0.55.$$

Thus

$$0.15 \leq P(A \cap B) \leq 0.55.$$

Every value in this interval gives four nonnegative probabilities summing to 1.

c. Under independence,

$$P(A \cap B) = 0.60(0.55) = 0.33$$

Question 41: Solution

$$P(A^c \cap B^c) = 1 - 0.60 - 0.55 + 0.33 = 0.18$$

$$P(\text{exactly one}) = (0.60 - 0.33) + (0.55 - 0.33) = 0.49$$

Check: $0.33 + 0.18 + 0.49 = 1$.

Question 42 (Average)

Sigma Hotels receives bookings through three channels. Each booking belongs to exactly one channel.

Channel	Current share of bookings	Cancellation probability within channel	Proposed share of bookings
App	0.50	0.04	0.20
Website	0.30	0.10	0.30
Agent	0.20	0.15	0.50

The proposed shares sum to 1. Assume the channel-specific cancellation probabilities would remain unchanged.

a. Calculate the overall cancellation probability under the current shares.

b. Calculate it under the proposed shares.

c. Quantify the change in percentage points. Would that change establish that cancellation behaviour had worsened within a channel?

Question 42: Solution

Let C denote cancellation and A, W, G denote app, website, and agent bookings. The three channels are nonoverlapping and cover all bookings, so they form a partition.

Channel	Current contribution to cancellation probability	Proposed contribution to cancellation probability
App	$0.50(0.04) = 0.020$	$0.20(0.04) = 0.008$
Website	$0.30(0.10) = 0.030$	$0.30(0.10) = 0.030$
Agent	$0.20(0.15) = 0.030$	$0.50(0.15) = 0.075$
Total	0.080	0.113

a.

$$P(C) = P(A)P(C \mid A) + P(W)P(C \mid W) + P(G)P(C \mid G)$$

$$P(C) = 0.50(0.04) + 0.30(0.10) + 0.20(0.15) = 0.020 + 0.030 + 0.030 = 0.080$$

b.

$$P_{\text{proposed}}(C) = 0.20(0.04) + 0.30(0.10) + 0.50(0.15)$$

$$P_{\text{proposed}}(C) = 0.008 + 0.030 + 0.075 = 0.113$$

c.

$$\text{Change} = 11.3\% - 8.0\% = 3.3 \text{ percentage points}$$

Question 42: Solution

The increase comes from a larger share of agent bookings and a smaller share of app bookings. The within-channel probabilities have not changed. A higher overall rate does not, by itself, establish deterioration within any channel.

Question 43 (Average)

Tau Payments uses a primary connection and a backup connection. Transactions can be processed whenever at least one connection is working.

Connection	Probability of working during the period
Primary	0.90
Backup	0.80

Initially assume the working statuses are independent.

a. Calculate the probability that transactions can be processed and the probability that exactly one connection is working.

b. Given that transactions can be processed, calculate the probability that the primary is down and the backup is carrying the service.

c. During heat alerts, the two individual working probabilities remain as above, but the probability that both connections are down is 0.07. Calculate service availability during a heat alert and assess whether the independence assumption remains valid.

Question 43: Solution

Let R denote a working primary connection and B a working backup.

a. Under independence,

$$P(R^c \cap B^c) = 0.10(0.20) = 0.02$$

$$P(\text{service available}) = 1 - 0.02 = 0.98$$

Equivalently, service is available when $R \cup B$ occurs:

$$P(R \cup B) = P(R) + P(B) - P(R \cap B)$$

$$P(R \cup B) = 0.90 + 0.80 - 0.90(0.80) = 0.98$$

Exactly one connection works in either of two mutually exclusive cases: the primary alone works, or the backup alone works.

$$P(\text{exactly one working}) = P(R \cap B^c) + P(R^c \cap B)$$

$$P(\text{exactly one working}) = 0.90(0.20) + 0.10(0.80) = 0.18 + 0.08 = 0.26$$

b. A failed primary and working backup already imply that service is available.

$$R^c \cap B \subseteq R \cup B$$

$$P(R^c \cap B \mid R \cup B) = \frac{P(R^c \cap B)}{P(R \cup B)}$$

Question 43: Solution

$$P(R^c \cap B \mid R \cup B) = \frac{0.10(0.80)}{0.98} = \frac{0.08}{0.98} = \frac{4}{49} \approx 0.0816$$

c.

$$P(\text{service available} \mid \text{heat alert}) = 1 - 0.07 = 0.93$$

Independence would imply a joint failure probability of $0.10(0.20) = 0.02$, not 0.07. The statuses are not independent under the heat-alert information. Availability falls from 98% to 93%, despite unchanged individual working probabilities.

Question 44 (Average)

Upsilon Assembly receives components from three suppliers. Each component comes from exactly one supplier.

Supplier	Share of components	Defect probability
Alpha	0.60	0.02
Beta	0.30	0.03
Gamma	0.10	0.09

a. Calculate the overall defect probability.

b. Given that a randomly selected component is defective, calculate the probability of each supplier being its source and rank the suppliers.

c. Compare this ranking with the ranking by defect probability within each supplier. Explain the difference.

Question 44: Solution

Let D denote a defect.

a.

Supplier	Prior share	Defect probability within supplier	Joint probability of supplier and defect
Alpha	0.60	0.02	$0.60(0.02) = 0.012$
Beta	0.30	0.03	$0.30(0.03) = 0.009$
Gamma	0.10	0.09	$0.10(0.09) = 0.009$

$$P(D) = 0.012 + 0.009 + 0.009 = 0.030$$

b.

$$P(\text{Alpha} \mid D) = \frac{0.012}{0.030} = 0.40$$

$$P(\text{Beta} \mid D) = \frac{0.009}{0.030} = 0.30, \qquad P(\text{Gamma} \mid D) = \frac{0.009}{0.030} = 0.30$$

The ranking is Alpha first, with Beta and Gamma tied.

c. By within-supplier defect probability, the order is Gamma (9%), Beta (3%), then Alpha (2%).

Question 44: Solution

Alpha supplies the largest share of components, so it contributes the largest share of defective components despite its lower defect rate. The source probabilities depend on both supply shares and within-supplier defect probabilities.

Question 45 (Average)

Phi Commerce screens orders for fraud.

Quantity	Probability
An order is fraudulent	0.02
Flag given a fraudulent order	0.90
Flag given a legitimate order	0.06

a. Calculate the probability that an order is flagged and the probability that a flagged order is fraudulent.

b. Calculate the probability that an unflagged order is fraudulent.

c. In a different order population, the fraud probability is 0.10, while both conditional flag probabilities remain unchanged. Recalculate the probability of fraud given a flag. Explain why it changes.

Question 45: Solution

Let D denote fraud and F denote a flag.

For parts (a) and (b), each interior cell below is a joint probability in the full order population.

Actual status	Flagged (F)	Unflagged (F^c)	Total
Fraudulent (D)	$0.02(0.90) = 0.018$	$0.02(0.10) = 0.002$	0.02
Legitimate (D^c)	$0.98(0.06) = 0.0588$	$0.98(0.94) = 0.9212$	0.98
Total	0.0768	0.9232	1.00

a. Use the flagged column.

$$P(F) = P(D)P(F \mid D) + P(D^c)P(F \mid D^c)$$

$$P(F) = 0.02(0.90) + 0.98(0.06) = 0.018 + 0.0588 = 0.0768$$

$$P(D \mid F) = \frac{P(D \cap F)}{P(F)} = \frac{0.02(0.90)}{0.0768} = \frac{0.018}{0.0768} = 0.234375 \approx 0.2344$$

b. Use the unflagged column.

$$P(F^c) = 1 - 0.0768 = 0.9232$$

$$P(D \cap F^c) = 0.02(1 - 0.90) = 0.002$$

$$P(D \mid F^c) = \frac{P(D \cap F^c)}{P(F^c)} = \frac{0.002}{0.9232} \approx 0.0022$$

Question 45: Solution

An unflagged order can still be fraudulent; the conditional probability is about 0.22%.

c. With a fraud share of 0.10, the legitimate share is 0.90. Recalculate the flagged probabilities for this population.

$$P(F) = 0.10(0.90) + 0.90(0.06) = 0.09 + 0.054 = 0.144$$

$$P(D \mid F) = \frac{0.09}{0.144} = 0.625$$

The fraud probability among flagged orders rises from about 23.44% to 62.50%. The screening probabilities within each status group are unchanged, but fraud is more common in the population being screened.

Question 46 (Average)

Chi Logistics needs a probability of late arrival for shipments on a newly introduced route.

Assessment	Evidence and reasoning
Operations analyst: 0.10	Ten percent of 10,000 shipments were late on the old route last year. The new route uses a different port and has a shorter connection window.
Route manager: 0.25	Judgement based on recent port reports and experience with similar connection windows. No matching shipment dataset or record of previous forecast accuracy is supplied.

The director wants to adopt 0.10 solely because it comes from many recorded shipments, describing 0.25 as “only an opinion.”

What would you advise? Explain how each probability was assigned, what each assessment does and does not establish, and what evidence would support the number used for the new route. No calculations are required.

Question 46: Solution

The 0.10 assessment uses relative frequency from the old route. It is well supported as a description of those records, but the large record count does not establish that the old rate transfers to a different port and connection window.

The 0.25 assessment is subjective. Its attention to the new operating conditions is relevant, but the supplied information does not establish the quality of the judgement or the numerical value.

Neither number should be preferred solely because of its label or the old dataset's size. Seek records from the new port and comparable connection windows, together with the manager's specific reasons for the adjustment. Assess whether the historical conditions match the intended use.

If a decision is needed before comparable records exist, use an explicitly provisional assessment with its assumptions recorded. Check it against new-route outcomes as they accumulate. Different numerical recommendations can be defended here if their assumptions and supporting evidence are made clear.

Question 47 (Hard)

Psi Engineering forms a four-person project team from nine distinct employees.

Qualification	Employees
Senior	Alpha, Beta, Gamma, Delta
Junior	Epsilon, Zeta, Eta, Theta, Iota

The team must contain at least two seniors. Alpha and Beta cannot serve together.

A complete appointment names the four team members, one senior team member as lead, and a different team member as recorder. Any selected member other than the lead can be recorder.

a. Count the eligible four-person teams.

b. Count the complete appointments.

c. Compare the probability of obtaining a team with exactly three seniors under these two procedures:

- Select an eligible team uniformly, then choose its lead uniformly from its seniors and its recorder uniformly from the other members.
- Select uniformly from all complete appointments.

Are the two procedures equivalent for choosing the team's composition?

Question 47: Solution

a. The senior-count cases do not overlap.

Seniors	Juniors	Eligible senior groups	Junior groups	Eligible teams
2	2	$\binom{4}{2} - 1 = 5$	$\binom{5}{2} = 10$	$5(10) = 50$
3	1	$\binom{4}{3} - \binom{2}{1} = 2$	$\binom{5}{1} = 5$	$2(5) = 10$
4	0	0	1	0

For two seniors, exclude the pair Alpha–Beta. For three seniors, exclude the two triples containing both Alpha and Beta. A four-senior team necessarily includes both.

$$N_{\text{teams}} = 50 + 10 = 60$$

b. A team with k seniors permits k choices of lead and three choices of recorder.

$$N_{\text{appointments}} = 50(2)(3) + 10(3)(3) = 300 + 90 = 390$$

c. Under uniform selection of eligible teams,

$$P(\text{three seniors}) = \frac{10}{60} = \frac{1}{6} \approx 0.1667.$$

Under uniform selection of complete appointments,

$$P(\text{three seniors}) = \frac{10(3)(3)}{390} = \frac{90}{390} = \frac{3}{13} \approx 0.2308.$$

Question 47: Solution

The procedures are not equivalent. Each three-senior team generates nine appointments, whereas each two-senior team generates six. Uniform selection of appointments therefore gives more weight to each three-senior team.

Question 48 (Hard)

Omega Components applies two tests to the same component.

Actual component status	Population share	Probability Test 1 flags	Probability Test 2 flags
Defective	0.05	0.80	0.80
Acceptable	0.95	0.10	0.10

Within either actual-status group, the two test results are independent: knowing one test's result does not change the other test's flag probability within that group.

a. Calculate the probability that both tests flag a component and the probability that such a component is defective.

b. Calculate the probability that exactly one test flags and the probability of a defect given exactly one flag.

c. An analyst treats the two flags as independent without conditioning on actual status. Check that claim numerically.

Question 48: Solution

Let D denote a defect and F_1, F_2 the two flags.

a. Both tests flag.

Independence applies within each actual-status group:

$$P(F_1 \cap F_2 \mid D) = P(F_1 \mid D)P(F_2 \mid D)$$

$$P(F_1 \cap F_2 \mid D^c) = P(F_1 \mid D^c)P(F_2 \mid D^c)$$

Actual status	Population share	Both flag within this group	Share of all components with this status and both flags
Defective (D)	0.05	$0.80(0.80) = 0.64$	$0.05(0.64) = 0.032$
Acceptable (D^c)	0.95	$0.10(0.10) = 0.01$	$0.95(0.01) = 0.0095$
Total	1.00		0.0415

$$P(F_1 \cap F_2) = P(D)P(F_1 \cap F_2 \mid D) + P(D^c)P(F_1 \cap F_2 \mid D^c)$$

$$P(F_1 \cap F_2) = 0.032 + 0.0095 = 0.0415$$

$$P(D \mid F_1 \cap F_2) = \frac{P(D \cap F_1 \cap F_2)}{P(F_1 \cap F_2)}$$

$$P(D \mid F_1 \cap F_2) = \frac{P(D)P(F_1 \cap F_2 \mid D)}{P(F_1 \cap F_2)}$$

Question 48: Solution

$$P(D \mid F_1 \cap F_2) = \frac{0.05(0.64)}{0.0415} = \frac{0.032}{0.0415} = \frac{64}{83} \approx 0.7711$$

Of the population share 0.0415 receiving both flags, 0.032 consists of defective components.

b. Exactly one test flags.

$$E = (F_1 \cap F_2^c) \cup (F_1^c \cap F_2)$$

The two cases are mutually exclusive. Every probability in the following table is conditional on the status group.

Test outcome	Defective (D)	Acceptable (D^c)
Test 1 flags; Test 2 does not	$0.80(0.20) = 0.16$	$0.10(0.90) = 0.09$
Test 1 does not flag; Test 2 flags	$0.20(0.80) = 0.16$	$0.90(0.10) = 0.09$
Exactly one flag within the group	$0.16 + 0.16 = 0.32$	$0.09 + 0.09 = 0.18$

$$P(E \mid D) = P(F_1 \mid D)P(F_2^c \mid D) + P(F_1^c \mid D)P(F_2 \mid D) = 0.32$$

$$P(E \mid D^c) = P(F_1 \mid D^c)P(F_2^c \mid D^c) + P(F_1^c \mid D^c)P(F_2 \mid D^c) = 0.18$$

Question 48: Solution

Multiply by the group shares to obtain joint probabilities in the full population:

Actual status and exactly one flag	Joint probability
Defective and exactly one flag: $D \cap E$	$0.05(0.32) = 0.016$
Acceptable and exactly one flag: $D^c \cap E$	$0.95(0.18) = 0.171$
Exactly one flag, either status: E	$0.016 + 0.171 = 0.187$

$$P(E) = P(D)P(E \mid D) + P(D^c)P(E \mid D^c) = 0.187$$

$$P(D \mid E) = \frac{P(D \cap E)}{P(E)} = \frac{P(D)P(E \mid D)}{P(E)}$$

$$P(D \mid E) = \frac{0.05(0.32)}{0.187} = \frac{16}{187} \approx 0.0856$$

c. Independence in the full population.

$$P(F_i) = P(D)P(F_i \mid D) + P(D^c)P(F_i \mid D^c), \qquad i = 1, 2$$

$$P(F_1) = P(F_2) = 0.05(0.80) + 0.95(0.10) = 0.135$$

Independence check	Probability
Both flags, calculated in part (a): $P(F_1 \cap F_2)$	0.0415
Product required for independence: $P(F_1)P(F_2)$	$0.135^2 = 0.018225$

Question 48: Solution

$$P(F_1 \cap F_2) = 0.0415 \neq 0.018225 = P(F_1)P(F_2)$$

Equivalently,

$$P(F_2 \mid F_1) = \frac{P(F_1 \cap F_2)}{P(F_1)} = \frac{0.0415}{0.135} \approx 0.3074$$

$$P(F_2 \mid F_1) \neq P(F_2) = 0.135$$

The flags are not independent overall: observing the first flag raises the probability of a second flag from 13.5% to about 30.74%. Independence within status groups does not imply independence after the groups are combined.

Question 49 (Hard)

Alpha Marketplace considers reviewing every order flagged by its fraud system.

Quantity	Value
Orders in each period	10,000
Fraud probability in the usual period	0.03
Fraud probability in a low-fraud period	0.002
Flag probability given fraud, in either period	0.80
Flag probability given a legitimate order, in either period	0.05
Review cost per flagged order	12 MAD
Loss from each fraudulent order not reviewed	150 MAD

A review identifies the order's true status and prevents the entire fraud loss if the order is fraudulent. Legitimate orders cause no fraud loss. Unflagged orders are not reviewed, there is no review-capacity limit, and no other costs apply.

Recommend whether to review every flagged order in each period. Support the recommendation with expected counts and total costs compared with reviewing no orders. Also determine the review cost per flagged order at which the two policies have equal expected total cost in the usual period.

Question 49: Solution

The counts below are expected counts under the stated probabilities, not guaranteed outcomes.

Status and flag result	Usual period	Low-fraud period
Fraudulent and flagged	$10{,}000(0.03)(0.80) = 240$	$10{,}000(0.002)(0.80) = 16$
Fraudulent and unflagged	$300 - 240 = 60$	$20 - 16 = 4$
Legitimate and flagged	$9{,}700(0.05) = 485$	$9{,}980(0.05) = 499$
Legitimate and unflagged	$9{,}700 - 485 = 9{,}215$	$9{,}980 - 499 = 9{,}481$
Total flagged	$240 + 485 = 725$	$16 + 499 = 515$
Total fraudulent	300	20

For the usual period:

$$C_{\text{review flags}} = 725(12) + 60(150) = 8{,}700 + 9{,}000 = 17{,}700 \text{ MAD}$$

$$C_{\text{review none}} = 300(150) = 45{,}000 \text{ MAD}$$

$$\text{Expected saving} = 45{,}000 - 17{,}700 = 27{,}300 \text{ MAD}$$

For the low-fraud period:

$$C_{\text{review flags}} = 515(12) + 4(150) = 6{,}180 + 600 = 6{,}780 \text{ MAD}$$

$$C_{\text{review none}} = 20(150) = 3{,}000 \text{ MAD}$$

Question 49: Solution

$$\text{Extra expected cost} = 6{,}780 - 3{,}000 = 3{,}780 \text{ MAD}$$

Review every flag in the usual period, but not in the low-fraud period under the stated two policy choices and costs. The system's conditional flag probabilities have not changed; the fraud share among flags has:

$$P(\text{fraud} \mid \text{flag, usual}) = \frac{240}{725} \approx 0.3310$$

$$P(\text{fraud} \mid \text{flag, low-fraud}) = \frac{16}{515} \approx 0.0311$$

For the break-even review cost c in the usual period:

$$725c + 60(150) = 300(150)$$

$$725c = (300 - 60)(150) = 36{,}000$$

$$c = \frac{36{,}000}{725} \approx 49.66 \text{ MAD per flagged order}$$

The exact threshold is $36{,}000/725$. Review costs below it favour reviewing every flag; costs above it favour reviewing none.

Question 50 (Hard)

Beta Distribution has two batches of distinct components.

Batch	Defective	Acceptable	Total
Alpha	2	4	6
Gamma	1	3	4

Two sampling procedures are proposed:

- Procedure A: choose Alpha with probability 0.60 or Gamma with probability 0.40, then select two components uniformly without replacement from that chosen batch.
- Procedure B: pool all ten components, then select two uniformly without replacement from the pool.

Under Procedure A:

a. Calculate the probability that exactly one selected component is defective.

b. Given exactly one defect in the selected pair, calculate the probability that the pair came from Alpha.

c. Given that the selected pair contains at least one defect, calculate the probability that both components are defective.

Then calculate the probability of exactly one defect under Procedure B. A colleague argues that the procedures must give the same result because the batch-selection probabilities in A match the batches' shares of the ten components. Assess that argument.

Question 50: Solution

Let E denote exactly one defect, H at least one defect, and T two defects.

a. Within each batch, all unordered pairs are equally likely.

Pair count	Alpha	Gamma
All pairs	$\binom{6}{2} = 15$	$\binom{4}{2} = 6$
Exactly one defect (E)	$\binom{2}{1}\binom{4}{1} = 8$	$\binom{1}{1}\binom{3}{1} = 3$
Two defects (T)	$\binom{2}{2} = 1$	0
At least one defect (H)	$8 + 1 = 9$	$3 + 0 = 3$

Divide each count by the total number of pairs in its batch:

Conditional probability	Alpha	Gamma
Exactly one defect	$8/15$	$3/6$
Two defects	$1/15$	$0/6 = 0$
At least one defect	$9/15$	$3/6$

$$P_A(E) = 0.60\left(\frac{8}{15}\right) + 0.40\left(\frac{3}{6}\right) = 0.32 + 0.20 = 0.52$$

Question 50: Solution

b.

$$P_A(\text{Alpha} \mid E) = \frac{P_A(\text{Alpha})P_A(E \mid \text{Alpha})}{P_A(E)}$$

$$P_A(\text{Alpha} \mid E) = \frac{0.60(8/15)}{0.52} = \frac{0.32}{0.52} = \frac{8}{13} \approx 0.6154$$

c. Gamma cannot supply two defective components.

$$P_A(T) = 0.60\frac{\binom{2}{2}}{\binom{6}{2}} + 0.40(0) = 0.60\left(\frac{1}{15}\right) = 0.04$$

Exactly one defect and two defects are nonoverlapping and together make up at least one defect.

$$P_A(H) = P_A(E) + P_A(T) = 0.52 + 0.04 = 0.56$$

Since $T \subseteq H$,

$$P_A(T \mid H) = \frac{P_A(T)}{P_A(H)} = \frac{0.04}{0.56} = \frac{1}{14} \approx 0.0714$$

For Procedure B, the pool contains three defective and seven acceptable components.

$$P_B(E) = \frac{\binom{3}{1}\binom{7}{1}}{\binom{10}{2}} = \frac{3(7)}{45} = \frac{7}{15} \approx 0.4667$$

Question 50: Solution

The procedures are not equivalent. Procedure A always selects both components from the same batch. Procedure B also permits pairs drawn from different batches. Matching the batch shares for the initial selection does not make the possible pairs or their probabilities identical.